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Emission rates of intermodal rail/road and road-only transportation in Europe: A comprehensive simulation study

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ABSTRACT

Intermodal rail/road transportation combines advantages of both modes of transport and is often seen as an effective approach for reducing the environmental impact of freight transportation. This is because it is often expected that rail transportation emits less greenhouse gases than road transportation. However, the actual emissions of both modes of transport depend on various factors like vehicle type, traction type, fuel emission factors, payload utilization, slope profile or traffic conditions. Still, comprehensive experimental results for estimating emission rates from heavy and voluminous goods in large-scale transportation systems are hardly available so far. This study describes an intermodal rail/road network model that covers the majority of European countries. Using this network model, we estimate emission rates with a mesoscopic model within and between the considered countries by conducting a large-scale simulation of road-only transports and intermodal transports. We show that there are high variations of emission rates for both road-only transportation and intermodal rail/road transportation over the different transport relations in Europe. We found that intermodal routing is more eco-friendly than road-only routing for more than 90% of the simulated shipments. Again, this value varies strongly among country pairs.

1. Introduction

In 2015, a total of 4452 million tons of anthropogenic CO_2 equivalent greenhouse gases were emitted in the European Union (EU), see Eurostat (2018b). Around 20% of these emissions resulted from transportation (Eurostat, 2018b), where road transportation is the dominant mode in the freight sector. For example, 75.3% of the total inland freight ton-kilometers (ton-km) in the EU were transported by truck, 18.3% by train and 6.4% by inland waterways in 2015 (Eurostat, 2018a). Recent research and political initiatives that seek for a reduction of the human-made environmental impact therefore also focus on emissions from transportation, see e.g. McKinnon et al. (2012), Psaraftis and Kontovas (2016), or DIN EN 16258 (2012). The amount of greenhouse gases (GHG) emitted per ton-km are used as a typical measure for the environmental impact of freight transportation. Although greenhouse gases subsume various substances like carbon dioxide, methane, nitrous oxide, hydrofluorocarbons, perfluorocarbons and sulphur hexafluoride, their global warming potential can be expressed as carbon dioxide equivalents (CO_2e) and, therefore, the amount of CO_2e per ton-km is considered a valid indicator of the environmental performance of freight transportation. Shifting freight from road to rail is often seen as an effective option to improve this measure (e.g. Dekker et al., 2012; McKinnon et al., 2012). Thereby, an objective comparison of transport options requires to consider well-to-wheel emissions that include not just the actual emissions of the transport operation (so called tank-to-wheel emissions) but also the emissions caused by producing and providing the fuel or energy to the

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vehicle (so called well-to-tank emissions), see Hoffrichter et al. (2012), Moro and Lonza (2017).

Research in the field of environmental transportation focusses on the development of eco-oriented routing and transport flow models (e.g. Bauer et al., 2010; Behnke and Kirschstein, 2017; Bektaş and Laporte, 2011; Ehmke et al., 2016; Rudi et al., 2016). Such studies usually consider emission rates obtained from emission estimation models or they use aggregated empirical rates reported in the literature. Experimental results in such papers typically refer to artificial networks or to relatively small real networks that cover a specific region. A comprehensive overview of emission rates that are observed in large-scale transportation systems like the European rail/road system is not provided by these studies. With our paper, we fill this gap and provide a systematic and broad overview of emission rates from intermodal rail/road and road-only transportation in Europe. More precisely, we analyze intra- and inter-country freight shipments for 27 European countries. We conduct a comprehensive simulation study that respects country specific attributes like the electrification and density of the rail network and geographical characteristics. To this end, we (i) collect data of an intermodal rail network across the considered countries, based on the corridors from the Trans-European Transport Network policy (TEN-T), see European Commission (2018), (ii) adapt a suitable emission estimation model for rail and road transportation and (iii) present results from an extensive simulation study that covers numerous experimental settings. We focus on relatively large shipments of several truck loads of heavy and voluminous goods, as these are candidates for rail transportation. Next to the differentiation between heavy and voluminous goods, we analyze the impact of further vehicle characteristics, shipment characteristics, and route characteristics such as traction type, electric emission factors, cargo size, gradient or traffic conditions. To estimate emissions, we adapt the mesoscopic model from Kirschstein and Meisel (2015). This model can capture both road and rail transportation and allows us to test the relevance of the previously mentioned parameters for the European transport system. We also contribute to the improvement of this estimation model by presenting an alternative way of approximating the gradient of a route. With these contributions, we present a broad and holistic overview of intermodal rail/road emission rates for European countries. The results of this study can be used to further facilitate research in areas where emission rates serve as an input to environmentally oriented decision problems, such as location planning or routing problems. They might also be used for companies to benchmark their environmental performance. In addition, the derived model of the European rail and road network can be used for follow-up research in the field of intermodal transportation.

The paper is organized as follows. Section 2 formally describes the intermodal rail/road network and the used data. Section 3 briefly reviews literature on emission estimation models and provides details about the mesoscopic estimation model. Section 4 describes the simulation settings and Section 5 presents the obtained results. Section 6 concludes this paper. The paper is supplemented by an extensive appendix that provides look-up tables for emission rates of road, rail and intermodal rail/road transportation for all country combinations and simulation settings considered in this study.

2. Modelling the European rail/road network

2.1. Multilayer network structure for intermodal transportation

Intermodal networks can be formally modelled as multilayer networks (e.g. Southworth and Peterson, 2000). An example of such network models is the Geospatial Intermodal Freight Transportation (GIFT) model for the United States that is also implemented as a web-tool (WebGIFT, 2018). This model covers various modes (rail, road, water) and enables the exploration of tradeoffs between cost, time and environmental measures. A description of the model and case studies can be found in Hawker et al. (2010),Winebrake et al. (2008) and Comer et al. (2010). For our study, we define here a rail/road network $\mathcal{G}(\mathcal{N}, \mathcal{E})$, where nodes $\mathcal{N} = \mathcal{N}^{rail} \cup \mathcal{N}^{trans} \cup \mathcal{N}^{road} \cup \mathcal{N}^{cust}$ and edges $\mathcal{E} \subset \mathcal{N} \times \mathcal{N}$ form four interconnected layers with subgraphs $\mathcal{G}^{rail}$, $\mathcal{G}^{trans}$, $\mathcal{G}^{road}$ and $\mathcal{G}^{cust}$, see Fig. 1. Nodes on these four layers are: (i) customer nodes $\mathcal{N}^{cust}$ that are origins and destinations of shipments, (ii) nodes $\mathcal{N}^{road}$ that

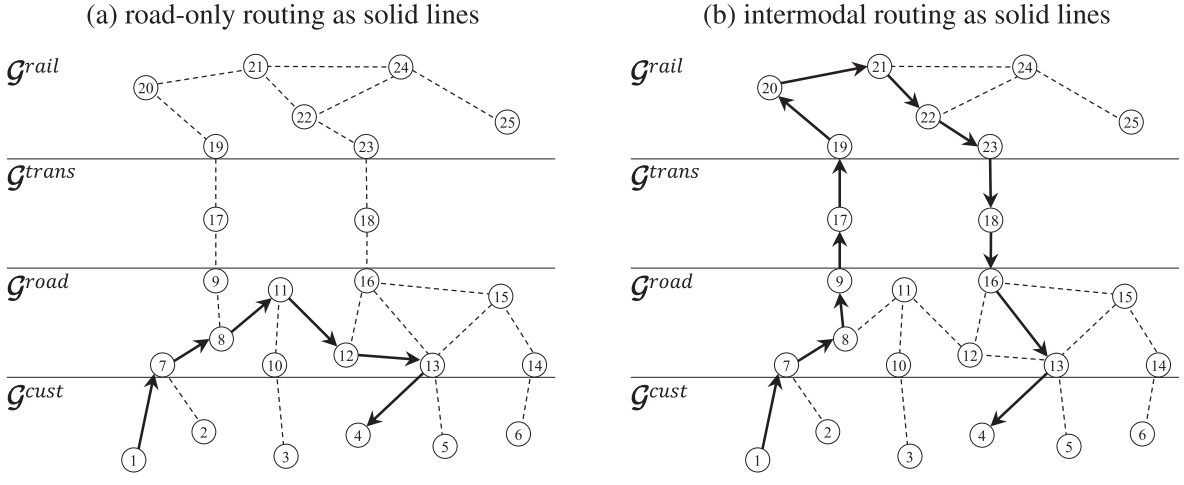


Fig. 1. Exemplary routings through an intermodal multilayer network.

belong to the road layer, (iii) nodes $\mathcal{N}^{trans}$ that enable freight transshipment between the road and rail mode, and (iv) nodes $\mathcal{N}^{rail}$ that belong to the rail layer.

The railway network $\mathcal{G}^{rail}(\mathcal{N}^{rail}, \mathcal{E}^{rail})$ within the uppermost layer is an incomplete network described by nodes $\mathcal{N}^{rail}$ (nodes 19–25 in the example) and edge set $\mathcal{E}^{rail} \subset \mathcal{N}^{rail} \times \mathcal{N}^{rail}$. The transshipment network $\mathcal{G}^{trans}(\mathcal{N}^{trans}, \mathcal{E}^{trans})$ represents the operational transshipment processes and consists of nodes $\mathcal{N}^{trans}$ (nodes 17 and 18 in the example) and an empty set $\mathcal{E}^{trans} = \emptyset$ as there are no edges between transshipment nodes. The road network $\mathcal{G}^{road}(\mathcal{N}^{road}, \mathcal{E}^{road})$ is an incomplete network described by nodes $\mathcal{N}^{road}$ (nodes 7-16 in the example) and edge set $\mathcal{E}^{road} \subset \mathcal{N}^{road} \times \mathcal{N}^{road}$. The bottom layer states the customer network $\mathcal{G}^{cust}(\mathcal{N}^{cust}, \mathcal{E}^{cust})$ and consists of all customer nodes $\mathcal{N}^{cust}$ (nodes 1-6 in the example) and an empty set $\mathcal{E}^{cust}$ as customers are only connected with each other via the road network. The additional cross-layer edge sets $\mathcal{E}^{trans-rail} \subset \mathcal{N}^{rail} \times \mathcal{N}^{trans}$, $\mathcal{E}^{trans-road} \subset \mathcal{N}^{trans} \times \mathcal{N}^{road}$ and $\mathcal{E}^{cust-road} \subset \mathcal{N}^{cust} \times \mathcal{N}^{road}$ connect subgraphs $\mathcal{G}^{rail}$, $\mathcal{G}^{trans}$, $\mathcal{G}^{road}$ and $\mathcal{G}^{cust}$. Railway nodes that qualify for transshipment to the road mode (nodes 19 and 23 in the example) are linked via dedicated transshipment nodes (nodes 17 and 18 in the example) to dedicated road nodes (nodes 9 and 16 in the example). Note that each node in $\mathcal{N}^{rail}$ and $\mathcal{N}^{road}$ is linked to at most one node in $\mathcal{N}^{trans}$ and that each node in $\mathcal{N}^{cust}$ is linked to exactly one node in $\mathcal{N}^{road}$.

We denote by $(i, j) \in \mathcal{E}$ the edge that connects node $i \in \mathcal{N}$ and node $j \in \mathcal{N}$. We consider asymmetric edge weights because edge characteristics (like gradient i^{α} and distance d introduced later in Section 3) might differ between edges $(i, j) \in \mathcal{E}$ and $(j, i) \in \mathcal{E}$ in a real-world infrastructure network. Eventually, shipments from an origin $i \in \mathcal{N}^{cust}$ to a destination $j \in \mathcal{N}^{cust}$ may use a road-only routing (Fig. 1a) or an intermodal routing (Fig. 1b). The road-only routing uses merely the two bottom layers whereas the intermodal routing uses all four layers. In other words, we assume that customers do not have direct access to the rail network, although this could be included in the model without difficulty.

2.2. Network data for the European rail and road network

We instantiate the theoretical network model from Section 2.1 to represent the rail/road network of the current EU member states, excluding the island states Cyprus, Ireland and Malta but including the non-EU member states Norway and Switzerland.

The railway network $\mathcal{G}^{rail}$ orients along the core corridors of the TEN-T policy. The use of TEN-T core corridors comes for several reasons: (i) the corridors define all relevant infrastructural elements, (ii) they also define rail line specific restrictions (e.g. a lack of electrification) and (iii) it can be assumed that all core corridors are relevant for intermodal freight transportation. The set of railway nodes $\mathcal{N}^{rail}$ is derived from respective terminals in the TENtec Interactive Map (2018). All these nodes qualify for rail/road transshipment. TEN-T contains at most one terminal for most of the cities. For cities with more than one terminal, we included only the

Table 1
Descriptive network data for 27 European countries.

Country	Country code [-]	Transshipment nodes [-]	Rail line		Emission coefficient [gCO_2e/kWh]	Customer nodes [-]
			Electrified [km]	Not electrified [km]		
Austria	AUT	4	1103	66	322	226
Belgium	BEL	7	838	0	261	83
Bulgaria	BGR	4	1114	0	618	297
Croatia	HRV	2	378	0	487	153
Czech Republic	CZE	6	1216	75	657	212
Denmark	DNK	2	343	121	364	116
Estonia	EST	1	56	216	878	116
Finland	FIN	5	520	0	207	384
France	FRA	16	4513	0	100	1500
Germany	DEU	26	5176	524	599	956
Greece	GRC	2	435	408	732	353
Hungary	HUN	2	1276	0	383	248
Italy	ITA	22	3919	0	413	806
Latvia	LVA	2	45	374	1110	170
Lithuania	LTU	3	95	493	370	172
Luxembourg	LUX	1	40	0	508	7
Netherlands	NLD	9	585	0	555	92
Norway	NOR	1	169	0	9	528
Poland	POL	9	3686	100	937	839
Portugal	PRT	5	1064	0	372	251
Romania	ROU	9	1494	112	449	630
Slovakia	SVK	2	663	0	412	132
Slovenia	SVN	2	397	109	309	55
Spain	ESP	14	3911	1349	321	1370
Sweden	SWE	5	1374	0	45	661
Switzerland	CHE	1	509	0	29	108
United Kingdom	GBR	9	1227	494	593	663
	$\sum$	171	36147	4439		11,128

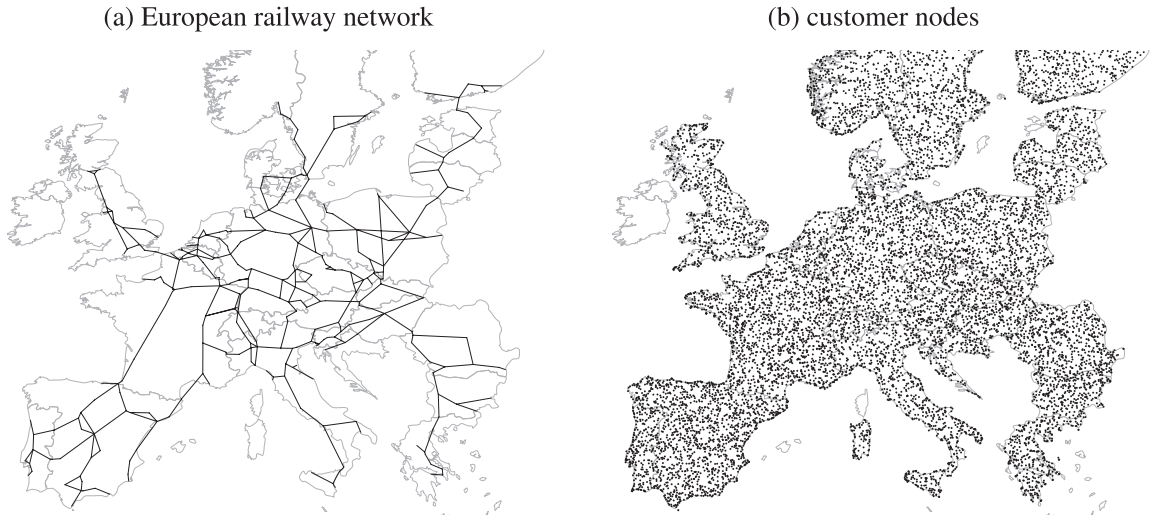


Fig. 2. Railway network and customer nodes.

most central one in our network. This results for example in 14 transshipments nodes for Spain and 4 transshipment nodes for Austria, see Table 1. Concerning railway edges $\mathcal{E}^{rail}$, we only consider existing tracks as derived from Raildar (2018). We supplement the rail network by auxiliary nodes that split some of the edges into two or more parts. This is to represent different edge characteristics, such as traction type or country borders. To increase the network connectivity, we include the projected 'Tallinn-Helsinki Tunnel' between Helsinki (Finland) and Tallinn (Estonia), the planned 'Fehmarn Belt Tunnel' between Puttgarden (Germany) and Rødby (Denmark), the train ferry that connects Sicily with mainland Italy and the train ferry that connects Rostock (Germany) with Trelleborg (Sweden), see European Parliament (2017), European Commission (2015), BluFerries (2018) and StenaLine (2018). The traction type of the locomotive on each railway track (either diesel or electric) is derived from information of the TEN-T Compliance Maps (2014). In total, $\mathcal{G}^{rail}$ includes 40,586 km of rail tracks, of which 36,147 km are electrified (89%) and 4439 km are not electrified (11%) and operated with diesel locomotives, see Table 1. Fig. 2a illustrates this railway network.

The road network $\mathcal{G}^{road}$ is derived from the Open Street Map (2018), as provided by Geofabrik (2018). The road network consists of all roads of the type "motorway", "trunk", "primary", "secondary", "tertiary" and "service". To improve the connectivity, we include heavy-good vehicle ferries, for example between Calais (France) and Dover (United Kingdom) as well as between Helsinki (Finland) and Stockholm (Sweden).

3. Emission estimation for road and rail transportation

3.1. Literature review

Numerous papers have investigated the environmental impact of transport activities and reported emission rates for various transport scenarios. Table 2 shows a selection of emission rates mentioned in the literature. Clearly, these values are valid only for the particular context considered in the respective study. In addition, some of the studies focus on carbon dioxide only and ignore other greenhouse gases and some of the studies look at well-to-wheel emissions whereas others consider tank-to-wheel emissions only. Therefore, the obtained emission rates might be taken up by other studies that demand emission rates as an input (for example to do eco-oriented transportation planning) but the comparability and explanatory power of the results is then limited. Therefore, it seems reasonable to provide a holistic overview of emission rates that covers a large scale transport system like the European rail/road network.

Generally, emissions caused by a particular transport activity can be computed using emission estimation models. In the literature, numerous models have been proposed that differ in the used data sources and the level of detail of the estimation. Psaraftis and Kontovas (2016) differentiate between top-down (or fuel-based) and bottom-up (or activity-based) methods. Top-down methods use the actual fuel consumption or estimate the fuel consumption of an already conducted transport. Emission factors for the respective fuel type are then used to quantify the GHG emissions. In contrast, bottom-up methods exploit relevant parameters of the transport activity (used vehicle type, distance, etc.) and estimate emissions with the help of look-up values. Another way to classify emission models is to differentiate between microscopic and macroscopic models (Scora and Barth, 2006). Microscopic models estimate emissions as precisely as possible by considering detailed physics of the moving vehicle, such as air resistance and rolling resistance. Macroscopic models are less precise and use, similar to bottom-up methods, empirical data to estimate emissions. Macroscopic models usually combine trip specific information, such as road profile and average speed, with vehicle specific information, such as tare weight and fuel type.

A large number of emission estimation models is designed either for road or for rail freight transportation. An overview of models

Table 2
Some emission rates per ton-km for road, rail and intermodal transportation.

Source	Emissions		Context
	Value	Unit	
Road			
de Miranda Pinto et al. (2017)	115	gCO_2	Case study for paper transportation in Brazil
Craig et al. (2013)	78	gCO_2	North America, varies between 55 and 106 gCO_2/ton-km
DEFRA (2017)	77	gCO_2e	HGV (> 33 t) in the UK, 50% load
NTM (2017b)	72	gCO_2e	EU average for a 34–40 t truck with trailer, 50% load
Winebrake et al. (2008)	48	gCO_2	Case study for the USA (13 t/TEU)
MMEEC (2018)	40	gCO_2	Generic truck with two TEU containers (13 t/TEU)
Rail			
DEFRA (2017)	34	gCO_2e	Freight train in the UK
NTM (2017a)	23	gCO_2e	EU average for diesel and electric locomotives
MMEEC (2018)	15	gCO_2	Generic line haul locomotive with 100 well cars (13 t/TEU)
Winebrake et al. (2008)	10	gCO_2	Case study for the USA (13 t/TEU)
Intermodal (rail/road)			
Craig et al. (2013)	42	gCO_2	North America, varies between 18 and 138 gCO_2/ton-km
de Miranda Pinto et al. (2017)	18	gCO_2	Case study for paper transportation in Brazil

for road transportation is provided by Demir et al. (2011). A popular model in this field is the Comprehensive Modal Emissions Model (CMEM) from Scora and Barth (2006). A popular model for rail transportation is the ARTEMIS (Assessment and Reliability of Transport Emission Models and Inventory Systems) model from Lindgreen and Sorenson (2005b). Both models follow the microscopic idea and derive emissions from a very detailed estimation of fuel or energy demands. Emission estimation models that cover both road and rail transportation often follow a macroscopic approach. The EcoTransIT World Initiative (2016) developed the EcoTransIT model that covers all major transportation modes (road, rail, water, air). Another macroscopic model for road and rail is the MEET (Methodologies for Estimating Air Pollutant Emissions from Transport) model from Hickman et al. (1999). The authors derive emission functions and parameters from real-world experiments for various vehicle types. Kirschstein and Meisel (2015) propose a mesoscopic model for road and rail transportation that combines the most relevant factors from microscopic models with empirical data from macroscopic models. The authors follow the physical base model from Ross (1997), which derives the required energy from factors like speed, weight, grade or acceleration. The relevance of these factors has been discussed in several other studies, such as Hickman et al. (1999) or Stead (1999). From the energy demand, the mesoscopic model deducts the amount of emitted greenhouse gases through emission factors. This procedure is similar to other emission estimation models, such as the model from DeLuchi (1991) or the GREET model from Wang (1999). Like Hickman et al. (1999) and others, the mesoscopic model supports trip specific acceleration profiles and thus allows for a consideration of various traffic conditions, like urban traffic and highway traffic. Kirschstein and Meisel (2015) show that this differentiation can have a significant impact on emission rates. Like all other models that cover road and rail, also the mesoscopic model can be used for an environmental assessment of intermodal rail/road transports. For the purpose of our study, the mesoscopic model describes a good trade-off between the required input data and the accuracy of estimated emissions. The next two subsections describe the methodology of the mesoscopic model for road transportation and for rail transportation.

3.2. Estimating emissions from road transportation

The mesoscopic model derives emissions for road transportation by estimating the fuel demand of a transport operation. Eq. (1) represents the main formula for computing GHG emissions and Table 3 shows the adopted notation. The formula first estimates the energy demand W_{truck} for moving the truck, taking into account the distance d of a considered trip, the gradient i of the route, the total weight m (including payload $m^{payload}$ and tare weight m^{tare}), the average number of acceleration processes per kilometer n^{acc} and the average speed v of the expected traffic conditions. For details of the corresponding computation, we refer to Kirschstein and Meisel (2015).

$$GHG_{truck} = \left(W_{truck}(d,\, i,\, m,\, n^{acc},\, v) \cdot \frac{r^{full} - r^{idle}}{\epsilon^t(v) \cdot p^{vehicle}} + r^{idle} \cdot \frac{d}{v} \right) \cdot k \tag{1}$$

Formula (1) weights W_{truck} to convert the energy demand into a fuel demand by taking into account the transmission efficiency of the truck $\epsilon^t(v)$, which depends on the average speed v, the maximal power of the engine $p^{vehicle}$ and the fuel consumption rate if the truck is idle r^{idle} or at full power r^{full}. The calculation requires further truck and fuel type specific parameters, namely the front surface A of the truck, the air resistance coefficient c^{air}, the rolling resistance coefficient c^{roll} and the energy coefficient p for diesel fuel. Corresponding values are also listed in Table 3. All values belong to a 40 t diesel truck, which is the vehicle type considered for road transportation throughout our study. Eventually, Formula (1) multiplies the fuel demand with a well-to-wheel emission coefficient k to convert the amount of diesel into an amount of CO_2e.

Table 3
Notation of the mesoscopic model for road transportation.

Notation	Value	Unit	Description
Trip specific			
W_{truck}	*	kWh	Required energy to move the truck
d	*	km	Distance
i	*	%	Gradient
m	*	t	Total weight
$m^{payload}$	*	t	Payload
n^{acc}	*	–	Average number of accelerations per km
v	*	km/h	Average speed
Truck and fuel type specific			
m^{tare}	14	t	Tare weight of the truck
$\epsilon^t(v)$	0.88	–	Transmission efficiency
$p^{vehicle}$	300	kW	Maximal power of the engine
r^{idle}	3	l/h	Fuel consumption rate if truck is idle
r^{full}	68.7	l/h	Fuel consumption rate if truck is at full power
A	9	m^2	Front surface area of the truck
c^{air}	0.6	–	Air resistance coefficient
c^{roll}	0.006	–	Rolling resistance coefficient
p	0.0811	l/kWh	Well-to-wheel energy coefficient (diesel)
k	3.15	$kgCO_2e/l$	Well-to-wheel emission coefficient (diesel)

* Computed from other parameters or varied in the simulation.

One of the parameters that needs to be provided to the mesoscopic model is the gradient i of the route. This gradient is then used to calculate the power to overcome the difference in altitude, which becomes part of the total energy to move the truck W_{truck}. For a path that connects some origin location with some destination location, an intuitive way to calculate this gradient i is to compare the altitude of the origin with the altitude of the destination. However, this approach can result in the same gradient value for two paths with very different slope profiles, see Fig. 3. The path from A to B in Fig. 3a goes through a hilly terrain with partially steep gradients whereas the path from C to D in Fig. 3b goes through a terrain with a modest but constant gradient. Since the altitude of origins A and C, the altitude of destinations B and D and the distance $d = 50$ km are identical, the simple origin-destination altitude difference leads to the same gradient for both routes and, thus, presumes an identical energy demand for both routes.

To resolve this issue, we introduce the concept of the adjusted gradient i^α, where we differentiate between positive and negative slopes. For this, we split the path, which is of total distance d, into n equally sized segments. We refer to the ith segment as z_i and denote by $alt(z_i^1)$ the altitude of the starting point of segment z_i and by $alt(z_i^2)$ the altitude of the ending point of segment z_i. Furthermore, we compute by $u_{z_i} = max\{alt(z_i^2) - alt(z_i^1), 0\}$ the positive altitude difference in segment z_i and by $v_{z_i} = max\{alt(z_i^1) - alt(z_i^2), 0\}$ the negative altitude difference in segment z_i. The calculation of the adjusted gradient i^α is shown in Formula (2) where $0 \leqslant \alpha \leqslant 1$ is a preset weight.

$$i^\alpha = \frac{1}{d} \cdot \sum_{z_i} (u_{z_i} - \alpha \cdot v_{z_i}) \tag{2}$$

The idea behind i^α is to fully account for positive slopes (u_{z_i}) and to (partially) subtract negative slopes (v_{z_i}). Note that negative slopes reduce the energy required to overcome other physical forces, such as rolling resistance or air resistance, and thus reduces the total energy demand (Ross, 1997). In addition, new technologies enable energy recoveries from negative slopes, such as the deceleration energy of trains. However, the original gradient i assumes that energy from negative slopes can be recovered totally. This might even lead to situations where energy recovered from negative slopes exceeds all other energy demands and results in a negative

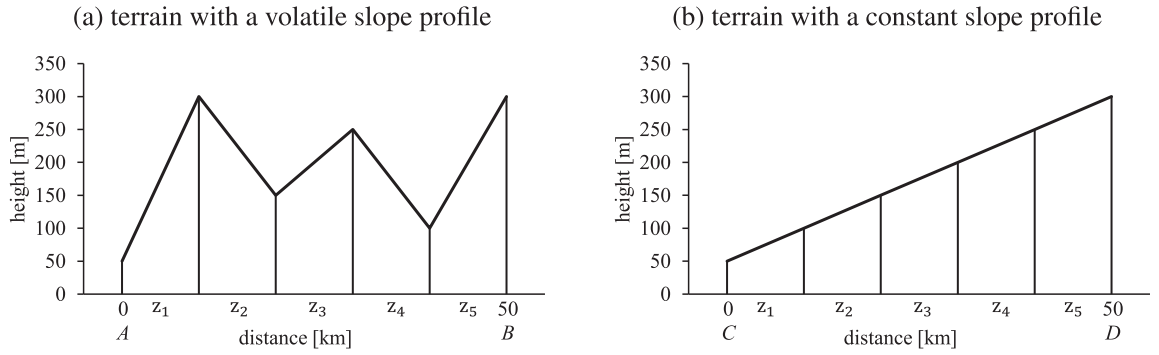


Fig. 3. Two routes with different slope profiles.

Table 4
Adjusted gradients i^{α} (in %) for the paths from Fig. 3.

	Fig. 3a		Fig. 3b	
α	$A \rightarrow B$	$B \rightarrow A$	$C \rightarrow D$	$D \rightarrow C$
0.0	1.10	0.60	0.50	0.00
0.2	0.98	0.38	0.50	−0.10
0.4	0.86	0.16	0.50	−0.20
0.6	0.74	−0.06	0.50	−0.30
0.8	0.62	−0.28	0.50	−0.40
1.0	0.50	−0.50	0.50	−0.50

overall energy demand, which is definitely unrealistic. With the introduction of α we can control to what extent this energy is recovered. In case $\alpha = 0$ only positive slopes are considered for the calculation of the adjusted gradient and downhill segments are treated like segments with flat terrain. In other words, $\alpha = 0$ means that no energy is recovered from negative slopes at all. With a value of $\alpha > 0$ energy from negative slopes is partially recovered. The extreme case of $\alpha = 1$ represents a full recovery of energy from negative slopes, identical to the original gradient i. In Table 4 we show i^{α} for the examples from Fig. 3 and for $\alpha \in [0, 0.2, \ldots, 1]$. The adjusted gradient in Fig. 3a increases with lower values of α, because the negative slopes are gradually ignored. For the reversed path that goes from B to A this leads to a change in the sign of i^{α}. In Fig. 3b, no segment of the path going from C to D has a negative altitude difference ($v_{z_i} = 0, \forall\, i = 1, \ldots, n$), which makes i^{α} independent of the value of α. Respectively, for the reversed path that goes from D to C, all segments have a negative altitude difference ($v_{z_i} > 0, \forall\, i = 1, \ldots, n$) and i^{α} decreases with higher values of α.

3.3. Estimating emissions from rail transportation

For rail freight transportation, the mesoscopic model differentiates between diesel and electric trains. Still, the main idea for both types of power supply is to estimate the energy demand W_{train} for moving the train and, then, either directly derive the emissions from this amount of energy (electric train) or use an energy coefficient p to first convert the energy demand into a fuel demand (diesel train). Formula (3) shows the calculation that is common for both types of power supply. Table 5 lists the corresponding notation.

$$GHG_{train} = \frac{W_{train}(d,\, i,\, m,\, n^{car},\, n^{acc},\, v)}{\epsilon} \cdot p \cdot k \tag{3}$$

The energy demand W_{train} is computed from distance d, gradient i, total weight m, number of rail cars n^{car}, number of accelerations per kilometer n^{acc} and average speed v. Here d, i, n^{acc} and v are defined identical to the road model. In contrast, m depends on the

Table 5
Notation of the mesoscopic model for rail transportation.

Notation	Value	Unit	Description
Trip specific			
W_{train}	*	kWh	Required energy to move the train
d	*	km	Distance
i	*	%	Gradient
m	*	t	Total weight
$m^{payload}$	*	t	Payload
n^{car}	*	–	Number of cars per train
n^{acc}	*	–	Average number of accelerations per km
v	*	km/h	Average speed
Train type specific			
m^{car}	17.7	t	Tare weight of a car
m^{loc}	123	t	Tare weight of the locomotive
ϵ	*	–	Efficiency of the locomotive
p	*	l/kWh	Energy coefficient
k	*	*	Well-to-wheel emission coefficient
A	10	m^2	Front surface area of the train
c_{car}^{air}	0.218	–	Air resistance car (3 transport units per car)
c_{loc}^{air}	1.1	–	Air resistance locomotive
$c_{aux_1}^{roll}$	0.0005	–	Rolling resistance (auxiliary 1)
$c_{aux_2}^{roll}$	0.0006	–	Rolling resistance (auxiliary 2)
c_{car}^{roll}	0.0006	–	Rolling resistance car
c_{loc}^{roll}	0.004	–	Rolling resistance locomotive

* Computed from other parameters or varied in the simulation.

payload $m^{payload}$, the number of rail cars n^{car} required for carrying this load, the tare weight per car m^{car} and the tare weight of the locomotive m^{loc}. The transmission efficiency of the locomotive ϵ, the energy coefficient p and the emission coefficient k depend on the type of power supply. For diesel trains, ϵ is set to 0.38, p is set to 0.0811 l/kWh and k is set to 3.15 $kgCO_2e$/l. For electric trains, ϵ is set to 0.65 and p is set to 1 (as it actually plays no role for electric trains). The emission coefficient k used by the electric train is measured in $kgCO_2e$/kWh and depends on the energy mix of the country within which the train operates. The calculation of W_{train} requires further train type specific parameters, which are listed in Table 5 but not described here in further detail. All mentioned values are for a train with a 6-axle locomotive, carrying a varying number of 19.64 m long four-axle 'Sgis' cars (Kirschstein and Meisel, 2015; Lindgreen and Sorenson, 2005a,b). These cars can carry at most three 20-foot containers or a total of 62 tons.

4. Simulation settings

Our simulation study aims at estimating emission rates for shipments within and between the considered European countries. Section 4.1 describes the creation of shipments and their routing in the network. Section 4.2 describes the specification of trip specific parameters for the mesoscopic model and derives an appropriate number of shipments to simulate per country pair. Finally, Section 4.3 describes different configurations that are later explored by experiments in Section 5.

4.1. Generation of shipments and routing

We generate shipments within and between 27 European countries. We denote by C the set of these countries. Given a country pair (c_1, c_2), with $c_1 \in C$ and $c_2 \in C$, we generate a set of shipments S_{c_1,c_2} for this transport relation. In case of $c_1 = c_2$ these shipments have their origin and destination in the same country whereas in case of $c_1 \neq c_2$ shipments have their origin and destination in different countries.

Each shipment $s \in S_{c_1,c_2}$ is defined by its payload l_s (in tons), its origin location o_s and its destination location d_s. For the payload l_s, we first draw a random shipment size $t_s \in [6, 7, ..., 24]$ of transport units (TU). Each such unit corresponds to a 20-ft equivalent container unit or a similar load device. Accordingly, the lowest shipment size of 6 transport units refers to 3 full truck loads or 2 full rail cars whereas the largest shipment size of 24 transport units refers to 12 full trucks or 8 full rail cars. We then assume a gross weight of 11 t per transport unit for heavy goods and 6 t per transport unit for voluminous goods, which leads to the payload l_s of shipment s. For the origin o_s and the destination d_s, we first preprocess a set of candidate customer node $\mathcal{N}_i^{cust}$ per country $i \in C$, with $\mathcal{N}^{cust} = \bigcup_{i \in C} \mathcal{N}_i^{cust}$. Node locations are drawn uniformly from latitude/longitude values that belong to the country, where we exclude locations that are not connected to the road network $\mathcal{G}^{road}$ (for example small islands). The number of candidate locations per country is proportional to the land area of this country. For France, which is the largest country in C, we draw 1,500 random locations. For each other country, the number of candidate locations derives from its land area in comparison to France. For example, the land area of Latvia is 11.3% of the area of France and we thus generate 170 customer nodes in Latvia. We exclude locations with a latitude north of 63° North, as these are sparsely populated areas in Norway, Sweden or Finland. Fig. 2b illustrates the nodes $\mathcal{N}^{cust}$ and Table 1 shows the number of customer nodes $|\mathcal{N}_i^{cust}|$ per country. For a particular shipment $s \in S_{c_1,c_2}$, we draw the origin o_s from set $\mathcal{N}_{c_1}^{cust}$ and the destination d_s from set $\mathcal{N}_{c_2}^{cust}$.

For each shipment $s \in S_{c_1,c_2}$, we compare two possible routings: road-only (RO) and intermodal (IM). For RO, shipment s goes from o_s to d_s solely through the road network $\mathcal{G}^{road}$. For IM, shipment s goes from o_s to a transshipment node $h_s^o \in \mathcal{N}^{trans}$ in the origin area via $\mathcal{G}^{road}$, uses the rail network $\mathcal{G}^{rail}$ for rail transportation to a transshipment node $h_s^d \in \mathcal{N}^{trans}$ in the destination area and finally goes from h_s^d to d_s via $\mathcal{G}^{road}$. We select h_s^o as the closest transshipment node to o_s and h_s^d as the closest transshipment node to d_s, where h_s^o is not necessarily within the same country as o_s and h_s^d is not necessarily within the same country as d_s. The IM routing of shipment s is therefore $o_s \rightarrow h_s^o$ via road, $h_s^o \rightarrow h_s^d$ via rail and $h_s^d \rightarrow d_s$ via road. For short distance shipments, this might lead to a selection of the same transshipment node $h_s^o = h_s^d$. In this case, the routing does actually not involve a rail transportation and we merely consider

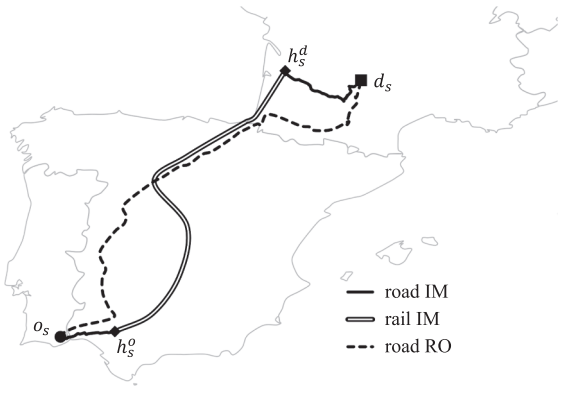


Fig. 4. Example of an intermodal and a road-only routing from Portugal to France.

road-only transportation for such shipments. Fig. 4 shows the RO and the IM routing for an exemplary shipment from Portugal to France. In the figure, the solid lines present the pre- and post-haulage $o_s \rightarrow h_s^o$ and $h_s^d \rightarrow d_s$ in the road mode, the two parallel solid lines represent the rail transportation $h_s^o \rightarrow h_s^d$ and the dashed line corresponds to the direct road transportation $o_s \rightarrow d_s$.

4.2. Specification of emission model parameters

We now specify the trip specific parameters distance d, gradient i, payload $m^{payload}$, number of rail cars n^{car}, average number of acceleration processes per kilometer n^{acc} and average speed v for a shipment s. We suppress here the shipment subscript s for reasons of readability. Furthermore, we specify the emission coefficient k for electric trains and derive an appropriate number of shipments to simulate for each pair of countries.

Distances d in the road-only mode and in the pre- and post-haulage of intermodal routings refer to the fastest routes in the road network $\mathcal{G}^{road}$ and are calculated with the Open Source Routing Machine (Luxen and Vetter, 2011). Distances d in the rail part of an intermodal routing refer to the shortest routes in the rail network $\mathcal{G}^{rail}$ and are calculated with Raildar (2018).

We approximate the gradients i^α in $\mathcal{G}^{road}$ and $\mathcal{G}^{rail}$ as follows. We consider the straight-line distance d_{o_s,d_s} from o_s to d_s and split this into segments of 10 km each. We use Open Elevation (2018) to obtain the altitude for each segment's start and end point. The gradient is then calculated from the slopes of all segments as described in Section 3.2.

For rail transportation, we assume a maximal total length of 740 meter per train (European Parliament and Council, 2013), which limits the number of cars per train to 36. For the calculation of the actual number of cars n^{car} and the payload $m^{payload}$, we assume a basic load of 24 cars, each holding 3 transport units à 9 ton. This load expresses a default train utilization rate of 67% w.r.t. the number of allowed cars per train. This rate is later varied in the experimental study. On top of that comes the shipment load l_s, which makes up the total payload of the train. The total emissions of the train on a specific edge are equally distributed over all cargo units in $m^{payload}$.

Kirschstein and Meisel (2015) showed that also the average number of acceleration processes per kilometer n^{acc} and the average speed v have a high influence on transport related emissions. We therefore use two settings for these values: one for congested traffic conditions and one for free-flowing traffic conditions, see Table 6. The parameter values for road transportation are based on real-world European driving cycles of high powered cars (André et al., 2006). For rail transportation, the average speed is based on real-world driving cycles of freight trains in Denmark (Lindgreen and Sorenson, 2005b). The average number of acceleration processes is set to 1 per 100 km for free-flowing traffic conditions and to 1 per 10 km for congested traffic conditions. Note that the average travel speed for rail transportation is rather high for congested traffic conditions. This reflects the freight trains' stop-and-go behavior, which is characterized by an acceleration to full speed between the stops.

For the well-to-wheel emission coefficient k for electric trains on edges $\mathcal{E}^{rail}$ that belongs to country $c_i \in C$, we use the energy mix of this particular country as reported in Moro and Lonza (2017), see Table 1. Finally, if ferry transportation is part of the routing, we use an emission rate of 51.35 gCO_2e/ton-km for this leg, see DEFRA (2017).

Having discussed the generation and routing of shipments, as well as the emission model parameters, we can now identify the number of shipments that needs to be simulated for a country pair (c_1, c_2) to obtain a valid estimate of the average emission rate. We derive this number from an analysis of the convergence index (Jung et al., 2004). The convergence index compares the deviation between the average rate observed after $n \in [1, 2, ..., N]$ simulated shipments with the average rate observed after simulating a total of N shipments. For this purpose, we simulate $N = 2000$ heavy goods shipments for country pairs (POL, PRT) and (FRA, FRA). For the first country pair, we expect similar routings for major parts of the journey and thus low variations in the rates. For the second country pair, we expect a high share of pre- and post carriages of variable length and thus higher variations in the rates. Fig. 5 illustrates the observed convergence of emission rates. The rates are higher for road only transportation than for intermodal transportation but converge in both cases quickly to the final rate observed after $N = 2000$ simulated shipments. For example, the intermodal emission rates for shipments within France are 20.06 gCO_2e/ton-km after simulating 500 shipments and 20.13 gCO_2e/ton-km after simulating 2000 shipments. For shipments between Poland and Portugal the rate is 16.64 gCO_2e/ton-km after 500 shipments and 16.62 gCO_2e/ton-km after 2000 shipments. From this analysis, we deduct that 500 shipments produce valid average emission rates and, therefore, we simulate $|S_{c_1,c_2}| = 500$ shipments for each country pair (c_1, c_2).

4.3. Definition of simulation configurations

We conduct our simulation study to calculate average emission rates for various relevant transport situations. We consider varied parameters for shipments (cargo type, size), varied parameters for the emission model (traffic conditions, gradient calculation) and

Table 6
Traffic conditions for road and rail transportation.

	Road		Rail	
	Congested	Free-flowing	Congested	Free-flowing
Average speed v [km/h]	40	80	70	80
Average accelerations n^{acc} [per km]	2.5	0.12	0.1	0.01

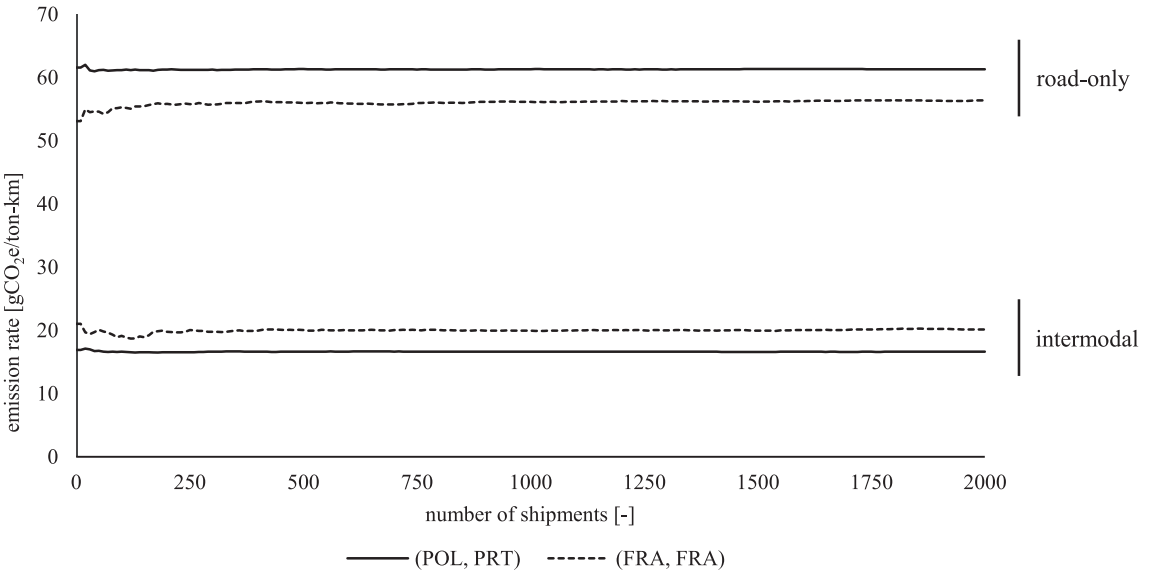


Fig. 5. Convergence of emission rates for country pairs (*POL*, *PRT*) and (*FRA*, *FRA*).

miscellaneous settings of additional parameters (train utilization rates, empty return trips, a green power scenario and emissions from transshipment operations). Table 7 provides an overview of all simulation settings. The default setting, configuration I, estimates emission rates with random shipment sizes between 6 and 24 transport units (TU), free-flowing traffic conditions, a value of $\alpha = 0.5$ for computing the adjusted gradient, a train utilization rate of 67%, no empty return trips, the country specific electric power mixes, the current degree of rail line electrification and no emissions from transshipment operations. Since we expect that the cargo weight has a major impact on the emission rates, we differentiate the cargo type in this configuration and all following configurations between heavy and voluminous goods.

In configuration II, we estimate emission rates with the assumption that all shipments have an identical size c_s of either 7 or 15 or 24 transport units. Configuration III analyses a setting where 50% of the distance in $\mathcal{G}^{road}$ and in $\mathcal{G}^{rail}$ is traveled under congested traffic conditions. Configuration IV looks at the impact of varying values for the parameter α, which determines the consideration of negative slopes in the calculation of the adjusted gradient i^{α}. Configuration V elaborates the impact of a train utilization rate of 50% and 78%, instead of the default 67%. Configuration VI includes empty vehicle trips in $\mathcal{G}^{road}$ and $\mathcal{G}^{rail}$ for the estimation of emissions from shipment *s*. With configuration VII we simulate the hypothetical case of a fully electrified rail network in Europe that is powered completely with 'clean' electricity. For this purpose, we apply the emission coefficient *k* from Sweden, which is the lowest *k* among all the EU member states, to the whole rail network. Finally, with configuration VIII, we consider emissions from transshipment processes in $\mathcal{G}^{trans}$ as it is a relevant issue in green intermodal transportation (Dekker et al., 2012; Marchant and Baker, 2012). Here, the amount of emitted greenhouse gases reflects the total energy demand of the terminal operations, such as crane liftings, warehousing or administrative actions, that is then allocated to single terminal tasks, see Clausen et al. (2013), Geerlings and Van Duin (2011) and Miodrag et al. (2016). In the study from Geerlings and Van Duin (2011), the authors estimate values of 14 and 23 $kgCO_2$ per container for two sea-rail transshipment terminals in the Port of Rotterdam and elaborate that terminal layout is important for an

Table 7
Configurations of simulation parameters.

	Configuration							
	I	II	III	IV	V	VI	VII	VIII
Shipment								
Cargo type* [–]	h/v	h/v	h/v	h/v	h/v	h/v	h/v	h/v
Shipment size t_s [TU]	[6–24]	**[7,15,24]**	[6–24]	[6–24]	[6–24]	[6–24]	[6–24]	[6–24]
Emission model								
Congested traffic [%]	0	0	**50**	0	0	0	0	0
Negative slopes α [–]	0.5	0.5	0.5	**[0–1]**	0.5	0.5	0.5	0.5
Miscellaneous								
Train utilization [%]	67	67	67	67	**[50,78]**	67	67	67
Empty return trips [–]	no	no	no	no	no	**yes**	no	no
Power mix & Electrification [–]	as is	as is	as is	as is	as is	as is	**green**	as is
Transshipment [$kgCO_2e$/TU]	0	0	0	0	0	0	0	**[1–25]**

* h: heavy, v: voluminous.

Table 8
Aggregated results for simulation configurations I to VIII.

	Emission rates for heavy goods [gCO_2e/ton-km]				Emission rates for voluminous goods [gCO_2e/ton-km]			
Config.	Road (RO)	Intermodal (IM)	Rail section	*IM-share*[*]	road (RO)	Intermodal (IM)	Rail section	*IM-share*[*]
I	56.8	23.8	18.1	92.7	86.7	29.2	19.4	94.9
II	[55.5–60.7]	[23.1–24.9]	[17.6–18.7]	[92.7–93.1]	[84.3–93.6]	[28.9–30.2]	[19.3–19.4]	[94.8–95.3]
III	64.8	25.7	19.0	93.5	95.6	31.2	20.2	95.2
IV	[48.8–64.9]	[17.7–30.0]	[12.5–23.8]	[89.6–94.6]	[75.9–97.4]	[22.5–36.0]	[13.5–25.3]	[93.8–95.7]
V	56.8	[23.6–24.3]	[17.9–18.7]	[92.3–92.8]	86.7	[28.8–30.1]	[19.0–20.4]	[94.7–95.0]
VI	92.6	32.3	21.9	94.7	152.2	44.7	26.4	95.7
VII	56.8	10.1	2.2	96.9	86.7	14.5	2.3	97.0
VIII	56.8	[23.8–26.6]	18.1	[89.1–92.7]	86.7	[29.2–34.4]	19.4	[91.7–94.9]

[*] Intermodal share (*IM-share*) in %, see Section 5.1.

accurate estimation. An estimation of transshipment emissions for a 40-foot container in Clausen et al. (2013) yields 6 $kgCO_2e$ for a transshipment from sea to land and 12 $kgCO_2e$ for a transshipment between rail and road. Since rail-road freight terminals can differ substantially in their layout, see Ballis and Golias (2002), we simulate in configuration VIII values between 1 and 25 $kgCO_2e$ per transport unit and per rail/road transshipment operation. With 27 countries considered in our study, 500 shipments per country pair, and a differentiation between heavy and voluminous goods, we simulate a total of 729,000 shipments for each of the eight configurations.

5. Results

Our results comprise estimated well-to-wheel emission rates in gCO_2e/ton-km for 729 country pairs. Emission rates reported for a country pair (c_1, c_2) are averages over the shipments $s \in S_{c_1,c_2}$. For the notation of the country pairs we use the country codes as reported in Table 1. We discuss in the following the results of selected country pairs only and refer to the Appendix A for a complete overview of all obtained results. A comprehensive summary of the results is shown in Table 8. Details of the results for configuration I and II are described in Section 5.1. Section 5.2 provides insights into the mesoscopic emission estimation model, including results from configurations III and IV. Finally, Section 5.3 presents results from configurations V, VI, VII and VIII.

5.1. General findings

In this section, we look at the spread of emission rates for the 729 country pairs under configuration I. We analyze emission rates from heavy and voluminous goods and we compare rates from road-only routing with rates from intermodal routing. Furthermore, we analyze the impact of the shipment size using configuration II.

A comparison of the emission rates across all country pairs shows that there is a wide spread of emission rates. Fig. 6 shows the relative frequency of emission rates from heavy shipments (Fig. 6a) and voluminous shipments (Fig. 6b) over all 729 country pairs. In general, intermodal routing (black bars) leads to lower average rates than road-only routing (grey bars), which confirms the advantage of rail transportation with respect to eco-friendliness. This finding is in line with other studies like those in Table 2. From our extensive simulation, we observe that emission rates from IM routings are on average 58% below rates from RO routings for heavy goods. For voluminous goods this measure is 66%. Emission rates for IM routings of both heavy and voluminous goods are mostly within a range of 15–45 gCO_2e/ton-km. For RO routings, emissions for heavy goods are mostly between 50 and 70 gCO_2e/ton-km and for voluminous goods between 80 and 100 gCO_2e/ton-km. The distribution of RO emissions is positively skewed at rates of about 50

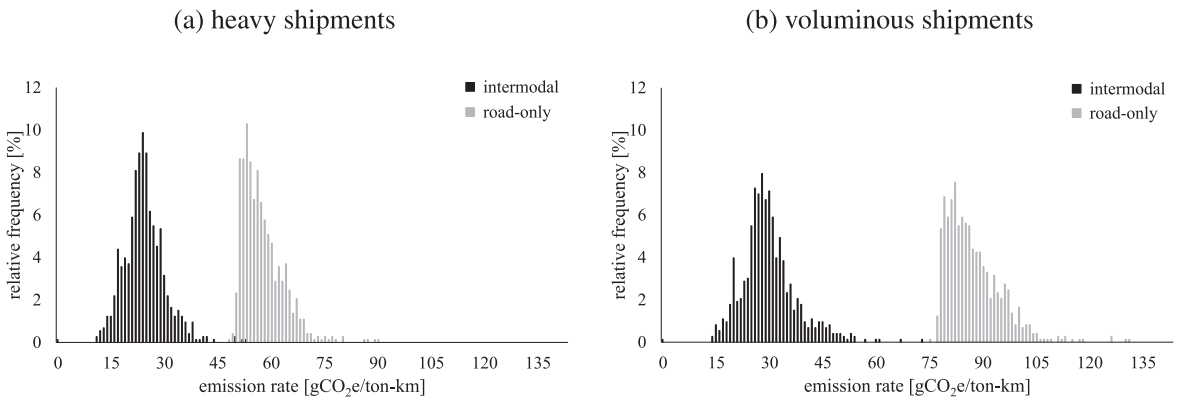


Fig. 6. Relative frequency of emission rates.

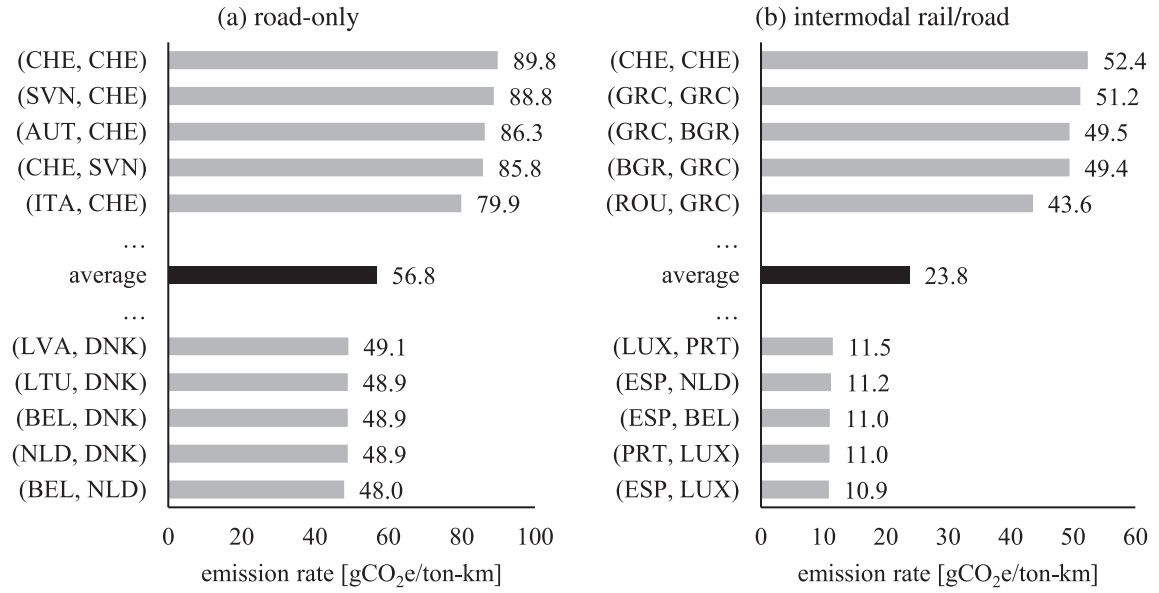


Fig. 7. Top 5 and bottom 5 emission rates for RO and IM heavy shipments.

for heavy shipments and about 80 gCO_2e/ton-km for voluminous shipments, see Fig. 6. This is because the gradient of a route is a major driver of RO emissions, where all country pairs with negligible altitude differences get close to these lower bounds. In contrast, for IM shipments, many more relevant drivers exist (traction type, power mix, etc.) which affects a more symmetric distribution of emission rates.

Fig. 7 shows the average and extreme emission rates from RO and IM routings of heavy goods under configuration I. The analysis shows that the extreme values for RO routing result from geographical characteristics, such as mountains. Routes with high emission rates pass the Alps, the highest mountain range in Europe, and routes with low emission rates typically involve flat terrain, as is found in Denmark, the Netherlands, Belgium, etc. In contrast, extreme values from IM routing result from the characteristics of a country's rail system, such as the share of electrification or the used power mix. High emissions are found for Greece, where just about half of the rail lines are electrified in combination with an unfavorable emission coefficient (see Table 1) that results from using lignite (brown coal) as a major source of energy (Public Power Corporation S.A., 2016). Low emissions are found for routes where substantial parts of the rail transportation goes through France, such as (*PRT*, *LUX*) or (*ESP*, *BEL*), as most of the rail network is electrified in combination with a low CO_2e-emission coefficient (see Table 1) that results from using nuclear power as a major source of energy (Réseau de Transport d'Électricité, 2017). The spread of emission rates for heavy RO transportation is between 48.0 and 89.8 gCO_2e/ton-km with an average of 56.8 gCO_2e/ton-km. For heavy IM transportation, the spread is between 10.9 and 52.4 gCO_2e/ton-km with an average of 23.8 gCO_2e/ton-km. These observations indicate that the use of a single average emission rate for a large area like Europe can lead to large errors in the estimation of emissions for a particular transport relation or a particular shipment.

The emission rates from shipments with voluminous goods are higher than the emission rates from shipments with heavy goods, see Table 8. For example, the average rate over all country pairs from RO routing is 56.8 gCO_2e/ton-km for heavy goods and 86.7 gCO_2e/ton-km for voluminous goods. For IM routing, the rate is 23.8 gCO_2e/ton-km for heavy goods and 29.2 gCO_2e/ton-km for voluminous goods. This finding shows that emission rates from road-only and intermodal routing are both sensitive to additional weight. This motivates presenting separate emission rates for heavy goods and for voluminous goods in all considered simulation configurations. We furthermore observe that the direction of a route can have an impact, i.e. that the results for country pair (c_1, c_2) and country pair (c_2, c_1) are not necessarily identical. However, such differences primarily exist for countries with significantly different average altitudes. For example, emission rates of shipments originating in Switzerland are usually lower than emission rates for shipments that go to Switzerland. This is hardly surprising as Switzerland is the country in Europe with the highest average elevation above sea level.

The comparison of emission rates per ton-km might lead to the conclusion that the intermodal routing of a shipment causes a lower amount of greenhouse gases than a road-only routing. However, this is too short sighted as the actual amount of emissions caused by a shipment also depends on the transport distances in the road and the rail mode, which can differ substantially in the RO and the IM routing. For the further analysis, we define the intermodal-share *IM-share* as the rate of those shipments where the intermodal route emits a lower amount of greenhouse gases than the road-only route. The average *IM-share* over all country pairs is 93% for heavy goods and 95% for voluminous goods. Around 80% of all country pairs have an *IM-share* of 90% or higher. These high values are because many shipments can use the rail mode for a substantial part of their routing. Likewise, short RO distances and high shares of pre- and post-carriages by truck characterize country pairs with low *IM-shares*. Examples are country pairs that include Finland, Norway or Sweden because the sparse rail network in these countries requires substantial pre- or post-carriages by truck. The *IM-share* is also low for country pairs with low electrification of the rail network or with high emission coefficients. For example, the

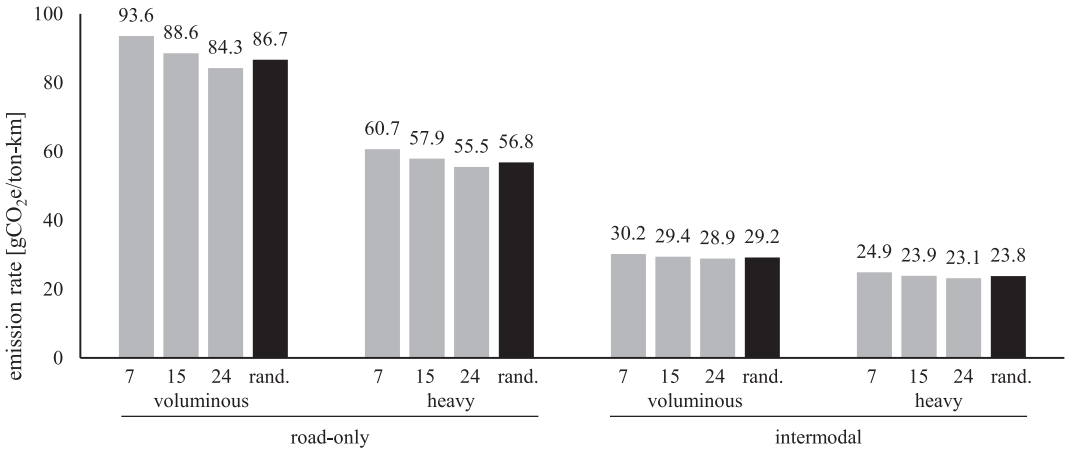


Fig. 8. Emission rates from fixed shipment sizes (grey bars) and random sizes (black bars).

IM-share for (*GRC*, *BGR*) is 48% for shipments with heavy goods and 79% for shipments with voluminous goods. This is because only half of the rail lines in Greece are electrified and both countries Greece and Bulgaria have a power mix with high emission coefficients, see Table 1.

The size t_s of a shipment s influences the number of trucks required, respectively the number of rail cars in the IM routing. Fig. 8 shows the average rates across all country pairs for shipments with random sizes (configuration I) and fixed sizes of 7 or 15 or 24 transport units (configuration II). The results show that the shipment size has only a small impact on the observed emission rates in RO routings. Since we consider quite large shipments of multiple transport units in all experiments, road-only emissions do not differ between even numbers of transport units and the only variance results from shipments with an uneven number of transport units. In this case, the relative impact of the emissions of the single half-filled truck depreciates with larger shipment size and the average emission rates tend towards the emission rates with even numbers of transport units. The impact of the shipment size is even lower in IM routings. The marginal emissions of an additional transport unit are very low, which is because of the high tare weight of trains and their substantial base utilization of 67%.

5.2. Emission model analysis

In this section, we look at the gradient of a route and the traffic conditions, which are two of the drivers of energy demand in the mesoscopic emission model. To further facilitate the understanding of transport related emissions, we analyze the composition of the total energy demand W. According to Kirschstein and Meisel (2015), W composes of energy to overcome air resistance, energy to overcome rolling resistance, energy to overcome grade and energy to perform accelerations, following the same physical laws for trucks and trains.

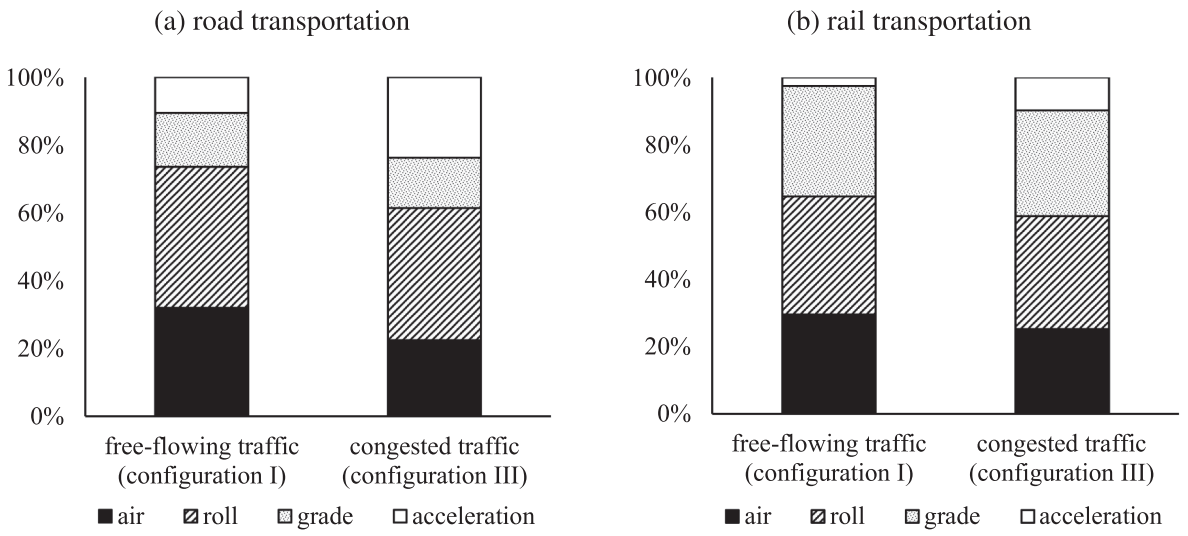


Fig. 9. Composition of the energy demand of heavy shipments.

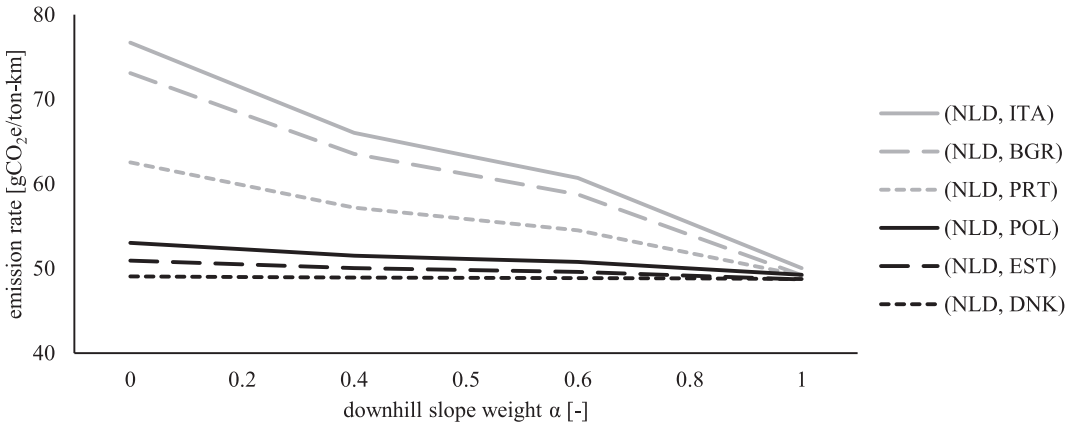


Fig. 10. Exemplary emission rates of heavy goods from configuration IV.

Fig. 9 shows the relative contribution of each of these parts to the total energy demand over all heavy goods shipments in configurations I and III. We see that the energy to overcome air resistance and rolling resistance accounts for more than 50% of the total energy demand, be it for trucks or for trains.

We observe that congested traffic has a stronger impact in RO routing than in IM routing, which is because the power to perform accelerations is more relevant in the road mode. The emission rates across all countries increase for RO routing by 14% (heavy) and by 10% (voluminous) and for IM routing by 8% (heavy) and by 7% (voluminous) when comparing free-flowing traffic and congested traffic, see Table 8. The impact of congested traffic conditions on the *IM-share* is 0.9% (heavy) and 0.3% (voluminous). We thus conclude that traffic conditions are important in estimating the height of emissions for a specific transport relation but they hardly determine which routing (RO or IM) is more eco-friendly.

With configuration IV, we look at the influence of considering energy from negative slopes through the average adjusted gradient i^{α}. For $\alpha = 0$, the adjusted gradient considers no energy from negative slopes at all. For $\alpha = 1$, where all energy from negative slopes is recovered, i^{α} reflects the simple origin-destination altitude difference and is identical to gradient i. For these extreme values, the average emission rate for heavy shipments over all country pairs is 17.7 ($\alpha = 1$) and 30.0 ($\alpha = 0$) for IM routings whereas it is 48.8 ($\alpha = 1$) and 64.9 ($\alpha = 0$) for RO routings. This supports the observation from Fig. 9 where the relative energy demand to overcome grade is higher for rail transportation than for road transportation. Clearly, these results vary among country pairs. Fig. 10 shows individual rates for six exemplary country pairs where shipments originate in the Netherlands with values of α being set to 0, 0.2, ...1. All country pairs have a similar emission rate for the case $\alpha = 1$, although their geographical characteristics are quite different. Shipments to Denmark, Estonia and Poland do not pass significant elevations, while shipments to Portugal, Bulgaria and Italy have to go through mountain ranges. Emission rates that are calculated with $\alpha = 1$ do not reflect these differences, whereas for lower values of α we see a clear spread in the emission rates that reflects these geographical characteristics. We thus conclude that the adjusted gradient i^{α} can better capture the characteristics of the diverse transport relations.

5.3. Further configurations

This section looks at the impact of varying train utilization (configuration V), empty returns trips (configuration VI), a fully electrified railway network (configuration VII) and emissions from transshipment operations (configuration VIII). For these configurations, we focus on discussing the aggregated results from Table 8. While configuration I assumes a train utilization rate of 67%, configuration V addresses utilization rates of 50% and 78%. We observe that a higher train utilization rate, expressed through a larger number of transport units and rail cars for the basic load, causes less marginal greenhouse gases from shipment load l_s and thus reduces the emission rate of shipment s. Likewise, a lower train utilization rate allocates more greenhouse gases to s. However, the absolute change is very low such that the average rate for heavy goods in IM routings varies between 23.6 for a train utilization of 78% and 24.3 gCO_2e/ton-km for a utilization of 50%. For voluminous goods the rate varies between 28.8 and 30.1 gCO_2e/ton-km. Since there is hardly any impact on the IM emission rate, there is also hardly any change on the *IM-share*.

The European Norm DIN EN 16258 (1625) states that emissions from transportation should also include vehicle operations from empty trips. Our results so far completely ignored empty trips. With configuration VI, we now consider this and assume empty vehicle return trips for all shipments. For this, we consider for a shipment s a corresponding shipment s' with $o_{s'} = d_s$ and $d_{s'} = o_s$ and load $l_{s'} = 0$. The IM routing for the return trip s' of shipment s is then d_s to h_s^d via road, h_s^d to h_s^o via rail and h_s^o to o_s via road. As expected, the consideration of empty return trips leads to a strong increase in the average emission rates of the freight shipments. The rate across all country pairs increases for RO routing by 63% (heavy) and by 75% (voluminous) and for IM routing by 36% (heavy) and by 53% (voluminous), respectively. Despite of this, we only observe a small increase in the *IM-share* by 2.0% (heavy) and 0.8% (voluminous), see Table 8. With configuration VII, we analyze the scenario of a fully electrified railway network. Furthermore, we apply a low emission coefficient of $k = 45$ gCO_2e/kWh to the whole network. This coefficient stems from Sweden, which is the EU member state with the lowest k. As expected, the emission rates from freight transportation on rail sections decrease for nearly all country

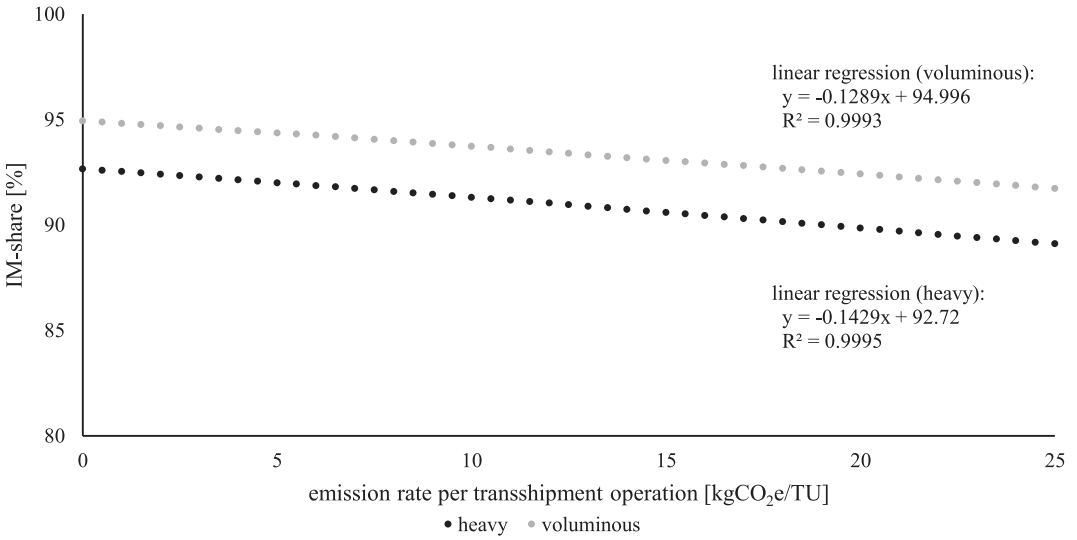


Fig. 11. *IM-share* depending on the emission rate per transshipment operation.

pairs with an average reduction of around 88%, see Table 8. This also leads to increasing *IM-shares*. Since the majority of country pairs already have high *IM-shares* in the base setting (configuration I) we see an increase of this measure of merely 4.2% for heavy goods and 2.1% for voluminous goods in configuration VII. Still, for some country pairs, we observe a much stronger impact of the green and fully electrified rail system. This is the case for countries that currently have high electric emission coefficients, like Latvia, Poland or Estonia. With a total of 438 billion ton-km of rail freight transport in the EU in 2016 (Eurostat, 2018c) and an average reduction of rail emission by 88% in a completely green rail network, the total amount of anthropogenic CO_2e emissions could be reduced by approximately 7.25 million tons. Finally, in configuration VIII, we consider emissions released during transshipment processes in the intermodal routing. Each of these routings includes two transshipment operations (one from road to rail and one from rail to road) where we assume an identical emission rate per transshipment operation. We varied this rate in the range 1–25 kgCO_2e per transshipped transport unit. Fig. 11 shows the average *IM-share* across all countries depending on the emission rate per transshipment operation. Surprisingly, we only observe a very small and constant decrease of about 0.1% per kgCO_2e added to the transshipment emission rate. However, we observe that the impact on the intermodal emission rates can be high for some country pairs. Here, the distance between shipments is the most important driver. This is because the emissions from transshipments are a fixed amount per shipment, which leads to a depreciation of these emissions with increasing distance. Fig. 12 illustrates this

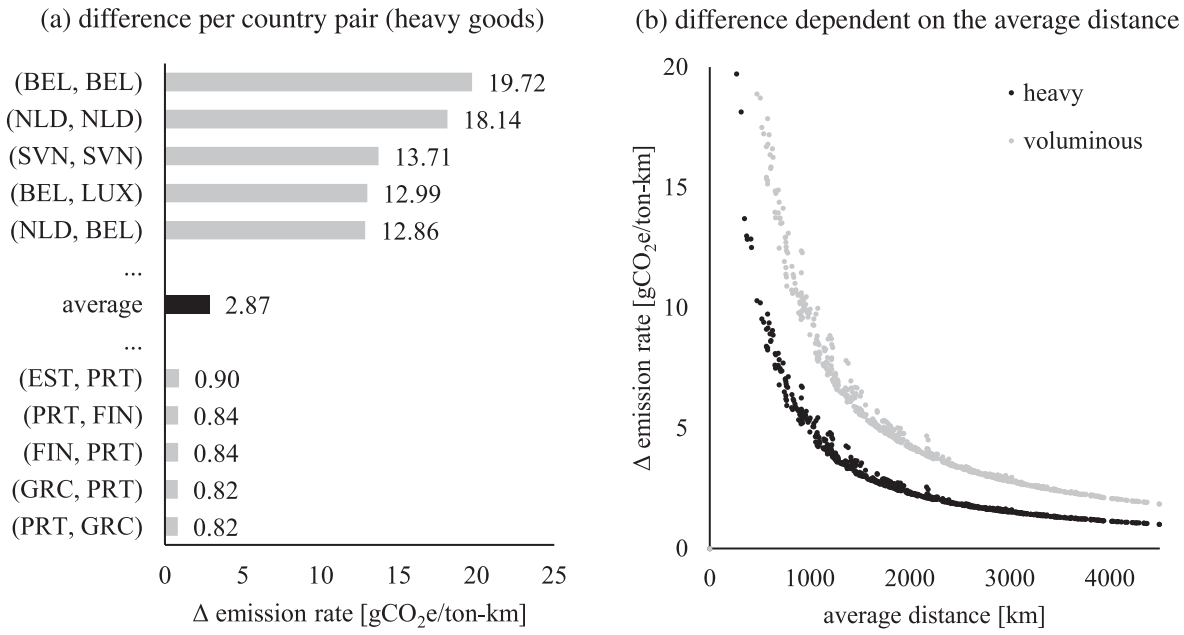


Fig. 12. Change of IM emissions rates for transshipment emissions of 25 kgCO_2e/TU.

relationship for a very high emission rate per transshipped TU of 25 kgCO_2e. It reports the increase in the intermodal emission rate per ton-km ($\triangle$ emission rate), when we compare configuration VIII with configuration I, i.e. if emissions from transshipments are added to the consideration. The five countries with the highest/lowest change in the emission rate per ton-km are shown in Fig. 12a and the relationship between the change of the emission rate per ton-km and the distance between country pairs is shown in Fig. 12b. Country pairs with low average distances, like (*BEL*, *BEL*) or (*NLD*, *NLD*), show the highest increase and country pairs with high average distances, like (*FIN*, *PRT*) or (*PRT*, *GRC*), show the lowest increase. The change in the emission rate is below 4 gCO_2e/ton-km for around 80% of all country pairs for heavy goods. For voluminous goods it is below 7 gCO_2e/ton-km for around 80% of all country pairs. Overall, we conclude that emissions from transshipment processes are relevant in estimating the height of emissions for a specific transport relation and in estimating the *IM-share*, especially for transport relations with low average distances, whereas we see hardly any impact for transport relations with high average distances.

6. Conclusion

In this paper, we have calculated emission rates of intermodal rail/road and road-only transportation of large cargo shipments in Europe. We have described the intermodal rail/road network, the various data sources and the results from a comprehensive simulation study. We found that the average emission rate across all country pairs from road-only routing is 57 gCO_2e/ton-km for heavy goods and 87 gCO_2e/ton-km for voluminous goods. For intermodal routing, we determined rates of 24 gCO_2e/ton-km for heavy goods and 29 gCO_2e/ton-km for voluminous goods. However, there are high variations. Rates for road transportation range from 48 to 131 gCO_2e/ton-km over the considered 729 country pairs. Rates for intermodal transportation range from 11 to 72 gCO_2e/ton-km. The highest emission rates are observed for transports that go through the Alps. Furthermore, we found that intermodal routings emit less greenhouse gases than road-only routings for over 90% of the simulated shipments. Again, this value varies among country pairs and is significantly lower for countries with many non-electrified rail tracks and high emission coefficients for their electricity. In our study, we have identified emission factors that quantify the value of road-only and intermodal emission rates and thus determine which routing is more eco-friendly. Some of these factors are of geographical or physical nature, such as gradient of the route or vehicle size, and thus rather hard or even impossible to change by decision makers from industry. Other factors, like average speed or the share of electrified rail lines, are set by transport operators or public authorities and thus could be addressed by policies that target a reduction of greenhouse gases from freight transportation.

Overall, our results indicate that the use of a single average emission rate for a large area like Europe can lead to substantial errors in the estimation of emissions for individual shipments or particular transport relations. The appendix of this study therefore provides look-up tables for all country pairs and shows the respective emission rates from road, rail and intermodal rail/road transportation as well as the *IM-share* for all considered scenarios. Future research should put a scope on smaller shipments and the impact of freight consolidation. It is also of interest to identify realistic values for the gradient parameter α and to validate our emission estimates through real-work experiment. Finally, we see the in-depth consideration of emissions from transshipment processes as another field of research for gaining deeper insights into the drivers of eco-friendly freight transportation.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.trd.2018.09.003.

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